



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The company experienced a DoS attack caused by an ICMP flood through an unconfigured firewall. Using the NIST CSF, the company should identify vulnerabilities, protect systems with firewalls and IDS/IPS, detect unusual traffic with monitoring tools, respond quickly by blocking malicious traffic, and recover critical services using backups and recovery plans.
Identify	Technology/Assets: Firewall, routers, servers, computers, and network services were affected. The attacker sent many ICMP pings through the unconfigured firewall and overwhelmed the network. Business Processes: Web design, graphic design, social media marketing, communication, and access to internal services were interrupted. People: Network administrators, cybersecurity staff, IT staff, and authorized employees need access to the affected systems.
Protect	Access Control: Allow only trusted traffic and users. Block unnecessary ICMP traffic and spoofed IP addresses. Training: Train IT and security staff about DoS attacks, firewall security, incident Data Security: Protect important company data with access controls, encryption, and backups.

	Procedures: Regularly review firewall rules and security procedures. Maintenance: Keep firewalls, operating systems, and security software updated. Protective Technology: Use firewalls, IDS/IPS, network monitoring, and ICMP rate limiting.
Detect	Anomalies: Use a SIEM to detect unusual traffic and security events. Monitoring: Continuously monitor network traffic, bandwidth, and firewall logs. Detection: Use IDS/IPS to detect and block suspicious ICMP traffic.
Respond	Response Planning: Create a DoS response plan with clear responsibilities. Communication: Quickly notify IT staff, management, and affected employees. Analysis: Check logs, identify the source of the attack, and determine affected systems. Mitigation: Block malicious traffic, isolate affected systems, and stop non-critical services. Improvements: Review the incident and improve firewall rules, monitoring, and response procedures.
Recover	Recovery Planning: Restore critical services first and then non-critical services. Improvements: Use backups, redundant systems, and regularly test recovery procedures. Communication: Inform employees and management about service restoration and when normal operations resume.

Reflections/Notes: This activity helped me understand how a DoS attack can affect a company's network and business operations. I learned how the NIST CSF can be used to identify risks, protect systems, detect attacks, respond to incidents, and recover services. I also learned that proper firewall configuration, network monitoring, and an incident response plan are important for preventing and reducing the impact of future attacks.

